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United States Patent

12415566

Kind Code

B2

Date of Patent

September 16, 2025

Inventor(s)

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Steering mechanism, vehicle, and apparatus and method for producing steering mechanism

Abstract

The present application provides a steering mechanism, a vehicle, and an apparatus and method for producing a steering mechanism, and belongs to the technical field of vehicles. Wherein the steering mechanism comprises a first directive wheel, a second directive wheel, a first driving device, and a second driving device. The first driving device is connected to the first directive wheel; and the second driving device is connected to the second directive wheel. The first driving device and the second driving device are configured to separately drive the first directive wheel and the second directive wheel respectively, thus making the first directive wheel and the second directive wheel deflect in the same direction to achieve steering. No linkage relationship exists between the two directive wheels, whereby the two directive wheels do not need to be connected through a complex mechanical transmission mechanism, and a structure of the steering mechanism is simplified.

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Appl. No.: 17/559784

Filed: December 22, 2021

Prior Publication Data

Document Identifier

Publication Date

US 20220144337 A1

May. 12, 2022

Related U.S. Application Data

continuation parent-doc WO PCT/CN2020/127375 20201107 PENDING child-doc US 17559784

Publication Classification

Int. Cl.: **B62D5/09** (20060101); **B62D5/06** (20060101); **B62D15/02** (20060101)

U.S. Cl.:

CPC **B62D5/09** (20130101); **B62D5/062** (20130101); **B62D15/024** (20130101);

Field of Classification Search

CPC: B62D (5/09); B62D (5/062); B62D (5/0418); B62D (5/12); B62D (15/024); B62D (9/00); B62D (7/00); B62D (7/09)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of International Application No. PCT/CN2020/127375, filed on Nov. 7, 2020, the disclosure of which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

(1) The present application relates to the technical field of vehicles, an in particular to a steering mechanism, a vehicle, and an apparatus and method for producing a steering mechanism.

BACKGROUND

(2) A steering mechanism of an existing vehicle is mainly divided into two types, the first type is a mechanical hydraulic steering mechanism, a hydraulic pump is driven through an engine belt, then pressure difference of hydraulic oil flowing into two ends of a power cylinder is controlled through a valve body, a piston is pulled to move, the piston is connected to a power connecting rod to transfer force to wheels to achieve left-right steering, and the mechanical hydraulic steering mechanism is mainly applied to a traditional fuel oil vehicle; and another type is that steering is controlled through electric power assistance, an electric machine, together with a clutch and a reduction gear, is installed on a frame, and the steering is achieved by assisting a mechanical power transmission rod by the motor.

(3) In general, linkage of two directive wheels in a steering mechanism is achieved through a complex linkage mechanism, and the whole steering mechanism structure is relatively complex.

SUMMARY

(4) Embodiments of the present application provide a steering mechanism, a vehicle, and an apparatus and method for producing the steering mechanism, thus improving the problem that the steering mechanism is complex in structure.

(5) In the first aspect, the embodiments of the present application provide a steering mechanism, comprising:

(6) a first directive wheel;

(7) a second directive wheel;

(8) a first driving device connected to the first directive wheel; and

(9) a second driving device connected to the second directive wheel;

(10) where the first driving device and the second driving device are configured to separately drive the first directive wheel and the second directive wheel respectively, thus making the first directive wheel and the second directive wheel deflect in the same direction to achieve steering.

(11) In the above solution, due to the fact that the first directive wheel and the second directive wheel are separately driven by the first driving device and the second driving device respectively to

deflect in the same direction, no linkage relationship exists between the two directive wheels, whereby the two directive wheels do not need to be connected through a complex mechanical transmission mechanism, and a structure of the steering mechanism is simplified.

(12) In addition, the steering mechanism provided by the embodiments of the present application also has the following additional technical features:

(13) In some embodiments, the first driving device includes:

(14) two first hydraulic cylinders distributed at the same side of the first directive wheel at interval, where one first hydraulic cylinder is configured to push the first directive wheel, and another first hydraulic cylinder is configured to pull the first directive wheel, thus driving the first directive wheel to deflect.

(15) In the above solution, the first directive wheel is driven to deflect by the two first hydraulic cylinders in a push-and-pull manner, the two first hydraulic cylinders may provide a larger steering force for the first directive wheel, and meanwhile the first directive wheel is more stable in the deflection steering process.

(16) In some embodiments, the second driving device includes:

(17) two second hydraulic cylinders distributed at the same side of the second directive wheel at interval, where one second hydraulic cylinder is configured to push the second directive wheel, and another second hydraulic cylinder is configured to pull the second directive wheel, thus driving the second directive wheel to deflect.

(18) In the above solution, the second directive wheel is driven to deflect by the two second hydraulic cylinders in a push-and-pull manner, the two second hydraulic cylinders may provide larger steering force for the second directive wheel, and meanwhile the second directive wheel is more stable in the deflection steering process.

(19) In some embodiments, the two first hydraulic cylinders include a first front hydraulic cylinder and a first rear hydraulic cylinder, the two second hydraulic cylinder includes a second front hydraulic cylinder and a second rear hydraulic cylinder, and the first front hydraulic cylinder, the first rear hydraulic cylinder, the second front hydraulic cylinder, and the second rear hydraulic cylinder are all located between the first directive wheel and the second directive wheel;

(20) when the first directive wheel and the second directive wheel deflect leftwards in the same direction, the first front hydraulic cylinder is configured to push the first directive wheel, the first rear hydraulic cylinder is configured to pull the first directive wheel, the second front hydraulic cylinder is configured to pull the second directive wheel, and the second rear hydraulic cylinder is configured to push the second directive wheel;

(21) when the first directive wheel and the second directive wheel deflect rightwards in the same direction, the first front hydraulic cylinder is configured to pull the first directive wheel, the first rear hydraulic cylinder is configured to push the first directive wheel, the second front hydraulic cylinder is configured to push the second directive wheel, and the second rear hydraulic cylinder is configured to pull the second directive wheel.

(22) In the above solution, due to the fact that the first front hydraulic cylinder, the first rear hydraulic cylinder, the second front hydraulic cylinder, and the second rear hydraulic cylinder are all located between the first directive wheel and the second directive wheel, the steering mechanism is compact in structure, and the occupied space is reduced.

(23) In some embodiments, the steering mechanism further includes:

(24) a hydraulic control system configured to supply liquid to the first driving device and the second driving device, thus making the first driving device and the second driving device separately drive the first directive wheel and the second directive wheel respectively, and making the first directive wheel and the second directive wheel deflect in the same direction.

(25) In the above solution, the liquid is supplied to the first driving device and the second driving device through the hydraulic control system, and thus the accurate control for controlling the first driving device and the second device may be achieved to make the first directive wheel and the

second directive wheel deflect in the same direction. It should be understood that the first driving device and the second driving device are hydraulic execution devices, and the first driving device and the second driving device may provide larger driving force for the first directive wheel and the second directive wheel under the action of the hydraulic control system.

(26) In some embodiments, the hydraulic control system includes:

(27) a steering control device having a first working state and a second working state;

(28) a steering part in transmission connection with the steering control device, where the steering part is configured to control the steering control device to switch between the first working state and the second working state; and

(29) a liquid supply device configured to supply liquid to the first driving device and the second driving device when the steering control device is in the first working state, thus making the first driving device and the second driving device respectively drive the first directive wheel and the second directive wheel to deflect leftwards in the same direction, and is also configured to supply liquid to the first driving device and the second driving device when the steering control device is in the second working state, thus making the first driving device and the second driving device respectively drive the first directive wheel and the second directive wheel to deflect rightwards in the same direction.

(30) In the above solution, the steering control device may be controlled by the steering part to make the steering control device be in the first working state or the second working state and make the liquid supply device supply liquid to the first driving device and the second driving device, and same-direction leftwards deflection or same-direction rightwards deflection of the first directive wheel and the second directive wheel is achieved. The hydraulic control system with such structure may achieve steering by operating the steering part, the overall structure is simple, and a steering operation is convenient.

(31) In some embodiments, the hydraulic control system further includes:

(32) a flow divider valve, where the liquid supplied by the liquid supply device flows into the first driving device and the second driving device through the flow divider valve and the steering control device in sequence; the flow divider valve is configured to regulate a proportion of flow of the liquid flowing into the first driving device from the steering control device to flow of the liquid flowing into the second driving device from the steering control device, thus making a deflection angle of the first directive wheel be the same as or different from a deflection angle of the second directive wheel.

(33) In the above solution, the proportion of the flow of the liquid flowing into the first driving device from the steering control device to the flow of the liquid flowing into the second driving device from the steering control device may be regulated through the flow divider valve. In an actual traveling process, the proportion of the flow of the liquid flowing into the first driving device to the flow of the liquid flowing into the second driving device may be controlled through the flow divider valve according to a specific driving condition, thus making the deflection angles of the first directive wheel and the second directive wheel in the steering process be the same or different, and the control performance and the driving feeling are improved.

(34) In some embodiments, the hydraulic control system further includes:

(35) a flow regulating valve, where the liquid supplied by the liquid supply device flows into the first driving device and the second driving device through the flow regulating valve, the flow divider valve, and the steering control device in sequence;

(36) a vehicle speed sensor for acquiring a vehicle speed signal of a vehicle; and

(37) a control unit for controlling the flow regulating valve to regulate the flow of the liquid flowing into the flow divider valve according to the vehicle speed signal.

(38) In the above solution, the vehicle speed signal of the vehicle may be acquired through the vehicle speed sensor, the flow regulating valve may be controlled according to the vehicle speed signal control unit to regulate the flow of the liquid flowing into the flow divider valve. If the flow

of the liquid flowing into the flow divider valve is changed, the flow of the liquid flowing into the first driving device and the second driving device through the flow divider valve and the steering control device in sequence is also changed, thus changing the driving force provided for the first directive wheel and the second directive wheel by the first driving device and the second driving device respectively.

(39) In some embodiments, the hydraulic control system further includes:

(40) a torque sensor for acquiring a torque signal of the steering part;

(41) and the control unit is also configured to control the flow regulating valve to regulate the flow of liquid flowing into the flow divider valve according to the torque signal.

(42) In the above solution, the torque signal of the steering part may be acquired through the torque sensor, and the flow regulating valve may be controlled according to the torque signal control unit to regulate the flow of the liquid flowing into the flow divider valve. If the flow of the liquid flowing into the flow divider valve is changed, the flow of the liquid flowing into the first driving device and the second driving device through the flow divider valve and the steering control device in sequence is also changed, thus changing the driving force provided for the first directive wheel and the second directive wheel by the first driving device and the second driving device respectively. When the torque applied to the steering part by a driver in the steering process of the steering part is greater than a preset value, the torque sensor may acquire a torque signal representing the magnitude of the torque applied to the steering part, and the control unit is configured to control the flow regulating valve to increase the flow of the liquid flowing into the flow divider valve, and the first driving device and the second driving device are ultimately configured to respectively provide larger driving force for the first directive wheel and the second directive wheel to make the first directive wheel and the second directive wheel deflect at a relatively high speed; when the torque applied to the steering part by a driver in the steering process of the steering piece is less than a preset value, the torque sensor may acquire a torque signal representing the magnitude of the torque applied to the steering part, and the flow regulating valve is controlled by the control unit to reduce the flow of the liquid flowing into the flow divider valve, and the first driving device and the second driving device are ultimately configured to respectively provide lesser driving force for the first directive wheel and the second directive wheel to make the first directive wheel and the second directive wheel deflect at a relatively slow speed.

(43) In some embodiments, the steering control device includes:

(44) a first reversing valve connected to the first driving device through a first pipeline and a second pipeline; and

(45) a second reversing valve communicated with the second driving device through a third pipeline and a fourth pipeline;

(46) where the steering part is configured to switch positions of a valve core of the first reversing valve and a valve core of the second reversing valve, thus making the steering control device be in a first working state or a second working state;

(47) when the steering control device is in the first working state, the first pipeline and the third pipeline are liquid inlet pipelines, and the second pipeline and the fourth pipeline are liquid return pipelines;

(48) and when the steering control device is in the second working state, the first pipeline and the third pipeline are liquid return pipelines, and the second pipeline and the fourth pipeline are liquid inlet pipelines.

(49) In the above solution, liquid inlet and outlet conditions of the first driving device and the second driving device are respectively changed through the first reversing valve and the second reversing valve, thus achieving the deflection of the first directive wheel and the second directive, and this steering control device is simple in structure.

(50) In some embodiments, the first driving device includes:

(51) two first hydraulic cylinders distributed at the same side of the first directive wheel at interval,

where a piston rod of each first hydraulic cylinder is connected to the first directive wheel, each first hydraulic cylinder includes a first cavity into which the liquid is fed to make the piston rod extend out and a second cavity into which the liquid is fed to make the piston rod retract;
(52) where the first cavity of one first hydraulic cylinder and the second cavity of another first hydraulic cylinder are communicated with the first pipeline, and the second cavity of the one first hydraulic cylinder and the first cavity of the another first hydraulic cylinder are communicated with the second pipeline.

(53) In the above solution, if the first pipeline is fed with liquid, the first cavity of one first hydraulic cylinder and the second cavity of another first hydraulic cylinder are fed with liquid, the piston rod of one first hydraulic cylinder extends out to apply thrust to the first directive wheel, and the piston rod of another first hydraulic cylinder retracts to apply tensile force to the first directive wheel; if the second pipeline is fed with liquid, the second cavity of one first hydraulic cylinder and the first hydraulic cylinder of another first hydraulic cylinder are fed with liquid, the piston rod of one first hydraulic cylinder retracts to apply tensile force on the first directive wheel, and the piston rod of another first hydraulic cylinder extends out to apply thrust to the first directive wheel. That is, whether the first pipeline or the second pipeline is fed with liquid, the two first hydraulic cylinders are configured to drive the first directive wheel to deflect in a push-and-pull manner, a larger steering force may be provided for the first directive wheel, and meanwhile, the first directive wheel is more stable in the deflection steering process.

(54) In some embodiments, the second driving device includes:

(55) two second hydraulic cylinders distributed at the same side of the second directive wheel at interval, where a piston rod of each second hydraulic cylinder is connected to the second directive wheel, each second hydraulic cylinder includes a third cavity into which the liquid is fed to make the piston rod extend out and a fourth cavity into which the liquid is fed to make the piston rod retract;

(56) where the third cavity of one second hydraulic cylinder and the fourth cavity of another second hydraulic cylinder are communicated with the third pipeline, and the fourth cavity of the one second hydraulic cylinder and the third cavity of another second hydraulic cylinder are communicated with the fourth pipeline.

(57) In the above solution, if the third pipeline is fed with liquid, the third cavity of one second hydraulic cylinder and the fourth cavity of another second hydraulic cylinder are fed with liquid, the piston rod of one second hydraulic cylinder extends out to apply thrust to the second directive wheel, and the piston rod of another second hydraulic cylinder retracts to apply tensile force to the second directive wheel; if the fourth pipeline is fed with liquid, the fourth cavity of one second hydraulic cylinder and the third hydraulic cylinder of another second hydraulic cylinder are fed with liquid, the piston rod of one second hydraulic cylinder retracts to apply tensile force on the second directive wheel, and the piston rod of another second hydraulic cylinder extends out to apply thrust to the second directive wheel. That is, whether the third pipeline or the fourth pipeline is fed with liquid, the two second hydraulic cylinders are configured to drive the second directive wheel to deflect in a push-and-pull manner, a larger steering force may be provided for the second directive wheel, and meanwhile, the second directive wheel is more stable in the deflection steering process.

(58) In a second aspect, the embodiments of the present application provide a vehicle, including the steering mechanism.

(59) In the above solution, the steering mechanism of the vehicle separately drives the first directive wheel and the second directive wheel to deflect in the same direction respectively through the first driving device and the second driving device, no linkage relationship exists between the two directive wheels, whereby the two directive wheels do not need to be connected through a complex mechanical transmission mechanism, a structure of the steering mechanism is simplified, and thus the total weight and the space occupation of the vehicle are reduced.

(60) In a third aspect, the embodiments of the present application provide an apparatus for producing a steering mechanism, where the apparatus for producing the steering mechanism includes:

(61) a providing device for providing a first directive wheel, a second directive wheel, a first driving device, and a second driving device; and

(62) an installing device for installing the first directive wheel and the second directive wheel on the first driving device and the second driving device respectively;

(63) where the first driving device and the second driving device are configured to separately drive the first directive wheel and the second directive wheel respectively, thus making the first directive wheel and the second directive wheel deflect in the same direction to achieve steering.

(64) In a fourth aspect, the embodiments of the present application provide a method for producing a steering mechanism, where the method for producing the steering mechanism includes:

(65) providing a first directive wheel, a second directive wheel, a first driving device, and a second driving device;

(66) installing the first directive wheel on the first driving device;

(67) and installing the second directive wheel on the second driving device;

(68) where the first driving device and the second driving device are configured to separately drive the first directive wheel and the second directive wheel respectively, thus making the first directive wheel and the second directive wheel deflect in the same direction to achieve steering.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) To describe the technical solutions in the embodiments of the present application more clearly, the following briefly introduces the accompanying drawings required for describing the embodiments of the present application. Apparently, the accompanying drawings in the following description show merely some embodiments of the present application, and those of ordinary skill in the art may still derive other drawings from these accompanying drawings without creative efforts.

(2) FIG. 1 is a structure diagram of a vehicle provided by some embodiments of the present application;

(3) FIG. 2 is a structure diagram of a steering mechanism (in which a first directive wheel and a second directive wheel are undeflected) provided by some embodiments of the present application;

(4) FIG. 3 is structure diagram of a steering mechanism (in which a first directive wheel and a second directive wheel are deflected leftwards) provided by some embodiments of the present application;

(5) FIG. 4 is a steering mechanism (in which a first directive wheel and a second directive wheel are deflected rightwards) provided by some embodiments of the present application;

(6) FIG. 5 is a structure diagram of a steering mechanism provided by one other embodiment of the present application;

(7) FIG. 6 is a structure diagram of a steering mechanism provided by some further embodiments of the present application;

(8) FIG. 7 is a structure diagram of a steering mechanism (in which a first directive wheel and a second directive wheel are undeflected) provided by yet further embodiments of the present application;

(9) FIG. 8 is a structure diagram of a steering mechanism (in which a first directive wheel and a second directive wheel are deflected leftwards) provided by yet further embodiments of the present application;

(10) FIG. 9 is a structure diagram of a steering mechanism (in which a first directive wheel and a

second directive wheel are deflected rightwards) provided by yet further embodiments of the present application;

(11) FIG. **10** is a structure diagram of a steering mechanism (in which a first directive wheel and a second directive wheel are undeflected) provided by some other embodiments of the present application;

(12) FIG. **11** is a structure diagram of a steering mechanism (in which a first directive wheel and a second directive wheel are deflected leftwards) provided by some other embodiments of the present application;

(13) FIG. **12** is a structure diagram of a steering mechanism (in which a first directive wheel and a second directive wheel are deflected rightwards) provided by some other embodiments of the present application;

(14) FIG. **13** is a structure diagram of a steering mechanism provided by some embodiments of the present application;

(15) FIG. **14** is a structure diagram of a steering mechanism provided by yet further embodiments of the present application;

(16) FIG. **15** is a schematic block diagram of an apparatus for producing a steering mechanism provided by some embodiments of the present application;

(17) FIG. **16** is a flow diagram of a method for producing a steering mechanism provided by some embodiments of the present application.

(18) Numeral description: **10**—first directive wheel, **11**—first wheel, **12**—first connecting frame, **13**—first rotating shaft, **20**—second directive wheel, **21**—second wheel, **22**—second connecting frame, **23**—second rotating shaft, **30**—first driving device, **31**—first hydraulic cylinder, **311**—first cavity, **312**—second cavity, **32**—first front hydraulic cylinder, **33**—first rear hydraulic cylinder, **40**—second driving device, **41**—second hydraulic cylinder, **411**—third cavity, **412**—fourth cavity, **42**—second front hydraulic cylinder; **43**—second rear hydraulic cylinder; **50**—hydraulic control system; **51**—steering control device; **511**—first reversing valve; **5111**—first pipeline; **5112**—second pipeline; **512**—second reversing valve; **5121**—third pipeline; **5122**—fourth pipeline; **52**—steering part; **53**—liquid supply device; **531**—hydraulic pump; **532**—liquid storage tank; **533**—driving motor; **54**—flow divider valve; **541**—pipe; **542**—connecting pipe; **55**—flow regulating valve; **56**—vehicle speed sensor; **57**—control unit; **58**—torque sensor; **60**—frame; **100**—steering mechanism; **200**—battery; **300**—controller; **400**—motor; **1000**—vehicle; **1100**—providing device; **1200**—installing device; **2000**—apparatus for producing steering mechanism.

DESCRIPTION OF EMBODIMENTS

(19) The embodiments of the present application are further described in detail below with reference to the accompanying drawings and the embodiments. The following detailed description of the embodiments and the accompanying drawings are provided to illustrate the principles of the present application and are not intended to limit the scope of the present application, that is, the present application is not limited to the described embodiments.

(20) In the description of the present application, it should be noted that, unless otherwise specified, “a plurality” means two or more; an orientation or positional relationship indicated by the terms “upper,” “lower,” “left,” “right,” “inner,” “outer,” and the like is merely for convenience in describing the present application and simplifying the description rather than indicating or implying that the referenced devices or elements must have a particular orientation, be constructed and operated in a particular orientation, and thus should not be construed as limiting the present application. In addition, the terms “first”, “second”, “third” and the like are used for descriptive purposes only and are not to be construed as indicating or implying relative importance. “Vertical” is not strictly vertical, but within the tolerance range. “Parallel” is not strictly parallel, but within the tolerance range.

(21) The directional terms in the following description are all directions shown in the drawings, and are not intended to limit the specific structure of the present application. In the description of the

present application, it also should to be noted that, unless otherwise explicitly specified or limited, the terms “installed,” “connected,” and “connection” are to be construed broadly, for example, either a fixed connection, a detachable connection, or an integral connection; direct connection or indirect connection through an intermediate. The specific meaning of the above terms in the present application may be understood as appropriate by those of ordinary skill in the art.

(22) At present, linkage of two directive wheels in a steering mechanism of a vehicle is generally achieved through a complex linkage mechanism, and the whole steering mechanism structure is relatively complex.

(23) In view of this, the present application provides a technical solution to achieve steering by separately driving two directive wheels to deflect in the same direction through two driving devices.

(24) It should be noted that the technical solutions described by the embodiments of the present application are all suitable for the vehicle, the vehicle may be a fuel oil vehicle, a gas vehicle, or a new energy vehicle, and the new energy vehicle may be a battery electric vehicle, a hybrid electric vehicle, or a range-extended electric vehicle. The embodiments of the present application are not particularly limited to the vehicle.

(25) For convenience of illustration, the following examples are illustrated with the vehicle as a new energy vehicle.

(26) Please refer to FIG. 1, FIG. 1 is a structure diagram of a vehicle **1000** provided by some embodiments of the present application, the vehicle **1000** includes a steering mechanism **100** and achieves steering through the steering mechanism **100**.

(27) In some embodiments, the vehicle **1000** further includes a battery **200**, and the battery **200** may be arranged at the bottom or head or tail of the vehicle **1000**. The battery **200** may be configured to supply power of the vehicle **1000**, for example, the battery **200** may serve as an operational power source of the vehicle **1000**.

(28) The vehicle **1000** may further include a controller **300** and a motor **400**; the controller **300** is configured to control the battery **200** to supply power for the motor **400**, for example, for operating power needs during startup, navigation, and driving of the vehicle **1000**.

(29) In some embodiments, the battery **200** may not only serve as an operational power source of the vehicle **1000**, but also as a driving power source of the vehicle **1000**, to provide driving power for the vehicle **1000** instead of or partially instead of fuel oil or natural gas.

(30) The battery **200** provided by the embodiments of the present application refers to a single physical module including one or more battery cells to provide higher voltage or capacity. For example, the battery **200** mentioned in the present application may include a battery module, or a battery pack and the like. The battery **200** generally includes a box body for packaging one or more battery cells, and the box body may prevent liquid or other foreign matters from affecting charging or discharging of the battery cell.

(31) In the present application, the battery cell may include a lithium-ion secondary battery, a lithium-ion primary battery, a lithium-sulfur battery, a sodium-lithium-ion battery, a sodium-ion battery, or a magnesium-ion battery and the like, which is not limited in the embodiments of the present application. The battery cell may be cylindrical, flat, rectangular, or other shapes, which is also not limited in the embodiments of the present application. In general, the battery cell is divided into three types according to a package mode: cylindrical battery cells, square battery cells, and soft package battery cells, which is also not limited in the embodiments of the application.

(32) The battery cell includes an electrode assembly and an electrolyte, and the electrode assembly is composed of a positive plate, a negative plate, and a separator. The battery cell works mainly on the movement of metal ions between the positive plate and the negative plate. The positive plate includes a positive current collector and a positive active material layer, the positive active material layer is coated on a surface of the positive current collector, and the current collector uncoated with the positive active material layer protrudes out of the current collector coated with the positive

active material layer, and the current collector uncoated with the positive electrode active material layer serves as a positive electrode tab. Taking the lithium ion battery as an example, the positive current collector may be made of aluminum, and the positive active material may be lithium cobalt oxide, lithium iron phosphate, ternary lithium, or lithium manganate. The negative plate includes a negative current collector and a negative active material layer, the negative active material layer is coated on a surface of the negative current collector, and the current collector uncoated with the negative active material layer protrudes out of the current collector coated with the negative active material layer, and the current collector uncoated with the negative electrode active material layer serves as a negative electrode tab. The negative current collector may be made of copper, and the negative active material may be carbon or silicon. To guarantee that fusing does not occur while a large current passes through, a plurality of positive electrode tabs are stacked together, and a plurality of negative electrode tabs are stacked together. The separator may be made of PP, or PE, and the like. In addition, the electrode assembly may be a wound structure or may be a laminated structure, and embodiments of the present application are not limited thereto.

(33) Please refer to FIG. 2 to FIG. 4, FIG. 2 is a structure diagram of a steering mechanism **100** (in which a first directive wheel **10** and a second directive wheel **20** are undeflected) provided by some embodiments of the present application, FIG. 3 is a structure diagram of a steering mechanism **100** (in which a first directive wheel **10** and a second directive wheel **20** are deflected leftwards) provided by some embodiments of the present application, and FIG. 4 is a structure diagram of a steering mechanism **100** (in which a first directive wheel **10** and a second directive wheel **20** are deflected rightwards) provided by some embodiments of the present application. The steering mechanism **100** provided by the embodiments of the present application includes a first directive wheel **10**, a second directive wheel **20**, a first driving device **30**, and a second driving device **40**. The first driving device **30** is connected to the first directive wheel **10**; and the second driving device **40** is connected to the second directive wheel **20**. The first driving device **30** and the second driving device **40** are configured to separately drive the first directive wheel **10** and the second directive wheel **20** respectively, thus making the first directive wheel **10** and the second directive wheel **20** deflect in the same direction to achieve steering.

(34) In the steering mechanism **100**, due to the fact that the first directive wheel **10** and the second directive wheel **20** are separately driven by the first driving device **30** and the second driving device **40** respectively to deflect in the same direction, no linkage relationship exists between the two directive wheels, whereby the two directive wheels do not need to be connected through a complex mechanical transmission mechanism, and a structure of the steering mechanism **100** is simplified.

(35) In addition, due to the fact that the first driving device **30** and the second driving device **40** may separately control the first directive wheel **10** and the second directive wheel **20** respectively, the first directive wheel **10** and the second directive wheel **20** are mutually independent and non-interference in the steering process; in the steering process, a deflection angle of the first directive wheel **10** driven by the first driving device **30** and a deflection angle of the second directive wheel **20** driven by the second driving device **40** may be the same or different, and the steering operability of the steering mechanism **100** is improved.

(36) The steering mechanism **100** may further include a frame **60**, the first driving device **30** and the second driving device **40** may be installed on the frame **60** to stably fix the first directive wheel **10** and the second directive wheel **20** to the frame **60**. In the steering process of the vehicle **1000**, the first directive wheel **10** and the second directive wheel **20** may be viewed as deflecting relative to the frame **60**. In the FIG. 2, FIG. 3, and FIG. 4, the first directive wheel **10** is located at the left side of the second directive wheel **20**.

(37) Where the first directive wheel **10** may include a first wheel **11** and a first connecting frame **12**; the first wheel **11** is rotatably arranged on the first connecting frame **12**, the first wheel **11** is able to rotate around its own axis relative to the first connecting frame **12**, the first driving device

30 is connected onto the first connecting frame **12**, and when the first driving device **30** drives the first connecting frame **12** to deflect, the first wheel **11** is to deflect along with the first connecting frame **12**, thus achieving deflection of the first directive wheel **10**.

(38) The second directive wheel **20** may include a second wheel **21**, and a second connecting frame **22**; the second wheel **21** is rotatably arranged on the first connecting frame **22**, the second wheel **21** is able to rotate around its own axis relative to the second connecting frame **22**, the second driving device **40** is connected onto the second connecting frame **22**, and when the second driving device **40** drives the second connecting frame **22** to deflect, the second wheel **21** is to deflect along with the first connecting frame **22**, thus achieving deflection of the second directive wheel **20**.

(39) In some embodiments, please continue to refer to FIG. 2 to FIG. 4, the first driving device **30** may include two first hydraulic cylinders **31** which are distributed at the same side of the first directive wheel **10** at interval, where one first hydraulic cylinder **31** is configured to push the first directive wheel **10**, and another first hydraulic cylinder **31** is configured to pull the first directive wheel **10**, thus driving the first directive wheel **10** to deflect.

(40) The first directive wheel **10** is driven to deflect by the two first hydraulic cylinders **31** in a push-and-pull manner, the two first hydraulic cylinders **31** may provide a larger steering force for the first directive wheel **10**, and thus the first directive wheel **10** is more stable in the deflection steering process.

(41) Where, for convenience in description, the two first hydraulic cylinders **31** are respectively defined as a first front hydraulic cylinder **32** and a first rear hydraulic cylinder **33**. It should be understood that the two first hydraulic cylinders **31** are arranged one behind the other, the first front hydraulic cylinder **32** is located at the front side of the first rear hydraulic cylinder **33**. The two first hydraulic cylinders **31** driving the first directive wheel **10** to deflect in a push-and-pull manner may be understood in that the first rear hydraulic cylinder **33** pulls the first directive wheel **10** when the first front hydraulic cylinder **32** pushes the first directive wheel **10** to deflect, and the first rear hydraulic cylinder **33** pushes the first directive wheel **10** when the first front hydraulic cylinder **32** pulls the first directive wheel **10** to deflect.

(42) A piston rod of each first hydraulic cylinder **31** is connected to the first directive wheel **10**, specifically, the piston rod of each first hydraulic cylinder **31** is hinged with the first connecting frame **12** of the first directive wheel **10**. Each first hydraulic cylinder **31** includes a first cavity **311** into which liquid is fed to make the piston rod extend out and a second cavity **312** into which liquid is fed to make the piston rod retract. If the first cavity **311** of the first front hydraulic cylinder **32** is fed with liquid, the piston rod of the first front hydraulic cylinder **32** extends out, and the first front hydraulic cylinder **32** pushes the first directive wheel **10** to deflect; if the second cavity **312** of the first front hydraulic cylinder **32** is fed with liquid, the piston rod of the first front hydraulic cylinder **32** retracts, and the first front hydraulic cylinder **32** pulls the first directive wheel **10** to deflect. Likewise, if the first cavity **311** of the first rear hydraulic cylinder **33** is fed with liquid, the piston rod of the first rear hydraulic cylinder **33** extends out, and the first rear hydraulic cylinder **33** pushes the first directive wheel **10** to deflect; and if the second cavity **312** of the first rear hydraulic cylinder **33** is fed with liquid, the piston rod of the first rear hydraulic cylinder **33** retracts, and the first rear hydraulic cylinder **33** pulls the first directive wheel **10** to deflect.

(43) In the embodiment, cylinder bodies of the two first hydraulic cylinders **31** may be directly fixed to the frame **60**, due to the fact that the deflection of the first directive wheel **10** is achieved through the two first hydraulic cylinders **31** in a push-and-pull manner, there may be no need to establish a rotatable connection relationship between the first directive wheel **10** and the frame **60**. Certainly, in other embodiments, please refer to FIG. 5, FIG. 5 is a structure diagram of a steering mechanism **100** provided by one other embodiment of the present application, the rotatable connection of the first connecting frame **12** of the first directive wheel **10** and the frame **60** may also be achieved through a first rotating shaft **13**, and in the steering process, the first directive wheel **10** is to rotate around the first rotating shaft **13**. In this case, the cylinder bodies of the two

first hydraulic cylinders **31** may be hinged with the frame **60**.

(44) It should be noted that, in other embodiments, other numbers of the first hydraulic cylinders **31** in the first driving device **30** may also be provided, for example, one first hydraulic cylinder **31** may be provided under the condition that the first connecting frame **12** of the first directive wheel **10** is rotationally connected to the frame **60** through the first rotating shaft **13**. Certainly, the first driving device **30** may be other structures, for example, the first driving device **30** is an electric machine for driving the first directive wheel **10** to deflect.

(45) In some embodiments, please refer to FIG. 2 to FIG. 4, the second driving device **40** may include two second hydraulic cylinders **41** which are distributed at the same side of the second directive wheel **20** at interval, where one second hydraulic cylinder **41** is configured to push the second directive wheel **20**, and another second hydraulic cylinder **41** is configured to pull the second directive wheel **20**, thus driving the second directive wheel **20** to deflect.

(46) The second directive wheel **20** is driven to deflect through the two second hydraulic cylinders **41** in a push-and-pull manner, the two second hydraulic cylinders **41** may provide a larger steering force for the second directive wheel **20**, and thus the second directive wheel **20** is more stable in the deflection steering process.

(47) Where, for convenience in description, the two second hydraulic cylinders **41** are respectively defined as a second front hydraulic cylinder **42** and a second rear hydraulic cylinder **43**. It should be understood that the two second hydraulic cylinders **41** are arranged one behind the other, the second front hydraulic cylinder **42** is located at the front side of the second rear hydraulic cylinder **43**. The two second hydraulic cylinders **41** driving the second directive wheel **20** to deflect in a push-and-pull manner may be understood in that the second rear hydraulic cylinder **43** pulls the second directive wheel **20** when the second front hydraulic cylinder **42** pushes the second directive wheel **20** to deflect, and the second rear hydraulic cylinder **43** pushes the second directive wheel **20** when the second front hydraulic cylinder **42** pulls the second directive wheel **20** to deflect.

(48) A piston rod of each second hydraulic cylinder **41** is connected to the second directive wheel **20**, specifically, the piston rod of each second hydraulic cylinder **41** is hinged with the second connecting frame **22** of the second directive wheel **20**. Each second hydraulic cylinder **41** includes a third cavity **411** into which liquid is fed to make the piston rod extend out and a fourth cavity **412** into which liquid is fed to make the piston rod retract. If the third cavity **411** of the second front hydraulic cylinder **42** is fed with liquid, the piston rod of the second front hydraulic cylinder **42** extends out, and the second front hydraulic cylinder **42** pushes the second directive wheel **20** to deflect; if the fourth cavity **412** of the second front hydraulic cylinder **42** is fed with liquid, the piston rod of the second front hydraulic cylinder **42** retracts, and the second front hydraulic cylinder **42** pulls the second directive wheel **20** to deflect. Likewise, if the third cavity **411** of the second rear hydraulic cylinder **43** is fed with liquid, the piston rod of the second rear hydraulic cylinder **43** extends out, and the second rear hydraulic cylinder **43** pushes the second directive wheel **20** to deflect; if the fourth cavity **412** of the second rear hydraulic cylinder **43** is fed with liquid, the piston rod of the second rear hydraulic cylinder **43** retracts, and the second front hydraulic cylinder **42** pulls the second directive wheel **20** to deflect.

(49) Preferably, the first front hydraulic cylinder **32**, the first rear hydraulic cylinder **33**, the second front hydraulic cylinder **42** and the second rear hydraulic cylinder **43** are all located between the first directive wheel **10** and the second directive wheel **20**, that is, the first front hydraulic cylinder **32** and the first rear hydraulic cylinder **33** are located at the inner side of the first directive wheel **10**, and the second front hydraulic cylinder **42** and the second rear hydraulic cylinder **43** are located at the inner side of the second directive wheel **20**. Such structure makes the steering mechanism **100** be compact in structure, and the occupied space is reduced.

(50) As shown in FIG. 3, when the first directive wheel **10** and the second directive wheel **20** deflect leftwards in the same direction, the first front hydraulic cylinder **32** is configured to push the first directive wheel **10**, the first rear hydraulic cylinder **33** is configured to pull the first

directive wheel **10**, the second front hydraulic cylinder **42** is configured to pull a second directive wheel **20**, and the second rear hydraulic cylinder **43** is configured to push the second directive wheel **20**.

(51) As shown in FIG. **4**, when the first directive wheel **10** and the second directive wheel **20** deflect rightwards in the same direction, the first front hydraulic cylinder **32** is configured to pull the first directive wheel **10**, the first rear hydraulic cylinder **33** is configured to push the first directive wheel **10**, the second front hydraulic cylinder **42** is configured to push a second directive wheel **20**, and the second rear hydraulic cylinder **43** is configured to pull the second directive wheel **20**.

(52) In the embodiment, cylinder bodies of the two second hydraulic cylinders **41** may be directly fixed to the frame **60**, due to the fact that the deflection of the second directive wheel **20** is achieved through the two second hydraulic cylinders **41** in a push-and-pull manner, there may be no need to establish a rotatable connection relationship between the second directive wheel **20** and the frame **60**. Certainly, in other embodiments, please continue to refer to FIG. **5**, the rotatable connection of the second connecting frame **22** of the second directive wheel **20** and the frame **60** may be achieved through a second rotating shaft **23**, and in the steering process, the second directive wheel **20** is to rotate around the second rotating shaft **23**. In this case, the cylinder bodies of the two second hydraulic cylinders **41** may be hinged with the frame **60**.

(53) It should be noted that, in still another embodiment, other numbers of the second hydraulic cylinders **41** in the second driving device **40** may also be provided, for example, one second hydraulic cylinder **41** may be provided under the condition that the second connecting frame **22** of the second directive wheel **20** is rotationally connected to the frame **60** through the second rotating shaft **23**. Certainly, the second driving device **40** may also be other structures, for example, the second driving device **40** is an electric machine for driving the second directive wheel **20** to deflect.

(54) In some embodiments, please refer to FIG. **6**, FIG. **6** is a structure diagram of a steering mechanism **100** provided by some further embodiments of the present application. The steering mechanism **100** may further include a hydraulic control system **50**, the hydraulic control system **50** is configured to supply liquid to the first driving device **30** and the second driving device **40**, thus making the first driving device **30** and the second driving device **40** separately drive the first directive wheel **10** and the second directive wheel **20** respectively, and thus making the first directive wheel **10** and the second directive wheel **20** deflect in the same direction to achieve steering.

(55) It should be understood that the first driving device **30** and the second driving device **40** are hydraulic execution devices, and the first driving device **30** and the second driving device **40** may provide a large driving force for the first directive wheel **10** and the second directive wheel **20** under the action of the hydraulic control system **50**. In various embodiments, the first front hydraulic cylinder **32**, the first rear hydraulic cylinder **33**, the second front hydraulic cylinder **42** and the second rear hydraulic cylinder **43** are all hydraulic execution devices. As an example, the hydraulic control system **50** supplies liquid to the first front hydraulic cylinder **32**, the first rear hydraulic cylinder **33**, the second front hydraulic cylinder **42**, and the second rear hydraulic cylinder **43**, a specific structure of the hydraulic control system **50** is set forth in detail below.

(56) In some embodiments, please refer to FIG. **7** to FIG. **9**, FIG. **7** is a structure diagram of a steering mechanism **100** (in which a first directive wheel **10** and a second directive wheel **20** are undeflected) provided by yet further some embodiments of the present application, FIG. **8** is a structure diagram of a steering mechanism **100** (in which a first directive wheel **10** and a second directive wheel **20** are deflected leftwards) provided by yet further embodiments of the present application, and FIG. **9** is a structure diagram of a steering mechanism **100** (in which a first directive wheel **10** and a second directive wheel **20** are deflected rightwards) provided by yet further embodiments of the present application. The hydraulic control system **50** may include a steering control device **51**, a steering part **52**, and a liquid supply device **53**.

(57) The steering control device **51** has a first working state and a second working state. The steering part **52** is in transmission connection with the steering control device **51**, and the steering part **52** is configured to control the steering control device **51** to switch between the first working state and the second working state.

(58) As shown in FIG. **8**, the liquid supply device **53** is configured to supply liquid to the first driving device **30** and the second driving device **40** when the steering control device **51** is in the first working state, thus making the first driving device **30** and the second driving device **40** respectively drive the first directive wheel **10** and the second directive wheel **20** to deflect leftwards in the same direction.

(59) As shown in FIG. **9**, the liquid supply device **53** is also configured to supply liquid to the first driving device **30** and the second driving device **40** when the steering control device **51** is in the second working state, thus making the first driving device **30** and the second driving device **40** respectively drive the first directive wheel **10** and the second directive wheel **20** to deflect rightwards in the same direction.

(60) The steering control device **51** may be controlled by the steering part **52** to make the steering control device **51** be in the first working state or the second working state and make the liquid supply device **53** supply liquid for the first driving device **30** and the second driving device **40**, and thus the same-direction leftwards deflection and same-direction rightwards deflection of the first directive wheel **10** and the second directive wheel **20** is achieved. The hydraulic control system **50** with such structure may achieve steering by operating the steering part **52**, the overall structure is simple, and the steering operation is convenient.

(61) The liquid supply device **53** may include a hydraulic pump **531** and a liquid storage tank **532**, the hydraulic pump **531** is configured to pump liquid from the liquid storage tank **532** into the steering control device **51**. The hydraulic pump **531** may be driven by a driving motor **533**. The liquid in the liquid storage tank **532** may be oil.

(62) The steering part **52** may be a steering wheel, and the switching of the steering control device **51** between the first working state and the second working state may be achieved by rotating the directive wheel.

(63) The steering control device **51** may have a variety of structures as long as it is possible to achieve steering of the first directive wheel **10** and the second directive wheel **20** by switching liquid paths.

(64) In some embodiments, please refer to FIG. **10** to FIG. **12**, FIG. **10** is a structure diagram of a steering mechanism **100** (in which a first directive wheel **10** and a second directive wheel **20** are undeflected) provided by some other embodiments of the present application, FIG. **11** is a structure diagram of a steering mechanism **100** (in which a first directive wheel **10** and a second directive wheel **20** are deflected leftwards) provided by some other embodiments of the present application, and FIG. **12** is a structure diagram of a steering mechanism **100** (in which a first directive wheel **10** and a second directive wheel **20** are deflected rightwards) provided by some other embodiments of the present application. The steering control device **51** may include a first reversing valve **511** and a second reversing valve **512**, where the first reversing valve **511** is connected to the first driving device **30** through a first pipeline **5111** and a second pipeline **5112**; and the second reversing valve **512** is connected to the second driving device **40** through a third pipeline **5121** and a fourth pipeline **5122**.

(65) The steering part **52** is configured to switch positions of a valve core of the first reversing valve **511** and a valve core of the second reversing valve **512**, thus making the steering control device **51** be in the first working state or the second working state.

(66) As shown in FIG. **11**, when the steering control device **51** is in the first working state, the first pipeline **5111** and the third pipeline **5121** are liquid inlet pipelines, and the second pipeline **5112** and the fourth pipeline **5122** are liquid return pipelines; as shown in FIG. **12**, when the steering control device **51** is in the second working state, the first pipeline **5111** and the third pipeline **5121**

are liquid return pipelines, and the second pipeline **5112** and the fourth pipeline **5122** are liquid inlet pipelines.

(67) Where the steering part **52** may be in transmission connection with the valve core of the first reversing valve **511** and the valve core of the second reversing valve **512** through a transmission mechanism to transfer a rotating force of the steering part **52** to the valve core of the first reversing valve **511** and the valve core of the second reversing valve **512**, thus making the valve core of the first reversing valve **511** and the valve core of the second reversing valve **512** move to change the working states of the first reversing valve **511** and the second reversing valve **512**. The transmission mechanism may have a variety of structural forms, such as, a pinion and rack mechanism.

(68) Optionally, a first cavity **311** of the first front hydraulic cylinder **32** and a second cavity **312** of the first rear hydraulic cylinder **33** are all communicated with the first pipeline **5111**, and the second cavity **312** of the first front hydraulic cylinder **32** and the first cavity **311** of the first rear hydraulic cylinder **33** are all communicated with the second pipeline **5112**. A third cavity **411** of the second front hydraulic cylinder **42** and a fourth cavity **412** of the second rear hydraulic cylinder **43** are all communicated with the fourth pipeline **5122**, and the fourth cavity **412** of the second front hydraulic cylinder **42** and the third cavity **411** of the second rear hydraulic cylinder **43** are all communicated with the third pipeline **5121**.

(69) As shown in FIG. **11**, when the steering control device **51** is in the first working state, due to the fact that the first pipeline **5111** and the third pipeline **5121** are liquid inlet pipelines, and the second pipeline **5112** and the fourth pipeline **5122** are liquid return pipelines, the first cavity **311** of the first front hydraulic cylinder **32** and the second cavity **312** of the first rear hydraulic cylinder **33** are fed with liquid through the first pipeline **5111**, the fourth cavity **412** of the second front hydraulic cylinder **42** and the third cavity **411** of the second rear hydraulic cylinder **43** are fed with liquid through the third pipeline **5121**, the liquid in the second cavity **312** of the first front hydraulic cylinder **32** and the liquid in the first cavity **311** of the first rear hydraulic cylinder **33** flow back through the second pipeline **5112** and finally flow back into the liquid storage tank **532**, and the liquid in the third cavity **411** of the second front hydraulic cylinder **42** and the liquid in the fourth cavity **412** of the second rear hydraulic cylinder **43** flow back through the second pipeline **5112** and finally flow back into the liquid storage tank **532**. During this process, the piston rod of the first front hydraulic cylinder **32** extends out to apply thrust to the first directive wheel **10**; the piston rod of the first rear hydraulic cylinder **33** retracts to apply tensile force to the first directive wheel **10**; the piston rod of the second front hydraulic cylinder **42** retracts to apply tensile force to the second directive wheel **20**; and the piston rod of the second rear hydraulic cylinder **43** extends out to apply thrust to the second directive wheel **20**. The same-direction leftward deflection of the first directive wheel **10** and the second directive wheel **20** is ultimately achieved.

(70) As shown in FIG. **12**, when the steering control device **51** is in the second working state, due to the fact that the first pipeline **5111** and the third pipeline **5121** are liquid return pipelines, and the second pipeline **5112** and the fourth pipeline **5122** are liquid inlet pipelines, the second cavity **312** of the first front hydraulic cylinder **32** and the first cavity **311** of the first rear hydraulic cylinder **33** are fed with liquid through the second pipeline **5112**; the third cavity **411** of the second front hydraulic cylinder **42** and the fourth cavity **412** of the second rear hydraulic cylinder **43** are fed with liquid through a fourth pipeline **5122**, the liquid in the first cavity **311** of the first front hydraulic cylinder **32** and the liquid in the second cavity **312** of the first rear hydraulic cylinder **33** flow back through the first pipeline **5111** and finally flow back into the liquid storage tank **532**, and the liquid in the fourth cavity **412** of the second front hydraulic cylinder **42** and the liquid in the third cavity **411** of the second rear hydraulic cylinder **43** flow back through the fourth pipeline **5122** and finally flow back into the liquid storage tank **532**. During this process, the piston rod of the first rear hydraulic cylinder **33** extends out to apply thrust to the first directive wheel **10**; the piston rod of the first front hydraulic cylinder **32** retracts to apply tensile force to the first directive

wheel **10**; the piston rod of the second rear hydraulic cylinder **43** retracts to apply tensile force to the second directive wheel **20**; and the piston rod of the second front hydraulic cylinder **42** extends out to apply thrust to the second directive wheel **20**. The same-direction rightward deflection of the first directive wheel **10** and the second directive wheel **20** is ultimately achieved.

(71) The first reversing valve **511** and the second reversing valve **512** may be three-position four-way directional control valves, the valve core of the first reversing valve **511** is provided with three positions: a left position, a right position, and a middle position. When the valve core of the first reversing valve **511** and the valve core of the second reversing valve **512** are both located at the left position, the steering control device **51** is in the first working state, the first pipeline **5111** supplies liquid to the first front hydraulic cylinder **32** and the first rear hydraulic cylinder **33**, the third pipeline **5121** supplies liquid to the second front hydraulic cylinder **42** and the second rear hydraulic cylinder **43**, and the first directive wheel **10** and the second directive wheel **20** deflect leftwards; when the valve core of the first reversing valve **511** and the valve core of the second reversing valve **512** are both located at the right position, the steering control device **51** is in the second working state, the second pipeline **5112** supplies liquid to the first front hydraulic cylinder **32** and the first rear hydraulic cylinder **33**, the fourth pipeline **5122** supplies liquid to the second front hydraulic cylinder **42** and the second rear hydraulic cylinder **43**, and the first directive wheel **10** and the second directive wheel **20** deflect rightwards. When the valve core of the first reversing valve **511** and the valve core of the second reversing valve **512** are both located at the middle position, none of the first pipeline **5111**, the second pipeline **5112**, the third pipeline **5121** and the fourth pipeline **5122** supplies the liquid, and thus none of the first directive wheel **10** and the second directive wheel **20** deflects.

(72) In other embodiments, the steering control device **51** may also be other structural valves, for example, the steering control device **51** is a rotary-actuated proportional valve, i.e., a proportional valve that is switched by rotation of the steering part **52**.

(73) In some embodiments, please refer to FIG. **13**, FIG. **13** is a structure diagram of a steering mechanism **100** provided by some embodiments of the present application. The hydraulic control system **50** further includes a flow divider valve **54**, where liquid supplied by the liquid supply device **53** flows into the first driving device **30** and the second driving device **40** through the flow divider valve **54** and the steering control device **51** in sequence; the flow divider valve **54** is configured to regulate the proportion of the flow of liquid flowing into the first driving device **30** from the steering control device **51** to the flow of liquid flowing into the second driving device **40** from the steering control device **51**, thus making the deflection angle of the first directive wheel **10** and the deflection angle of the second directive wheel **20** be the same or different.

(74) During actual traveling process, the proportion of the flow of the liquid flowing into the first driving device **30** to the flow of the liquid flowing into the second driving device **40** may be controlled through the flow divider valve **54** according to a specific driving condition to make the deflection angles of the first directive wheel **10** and the second directive wheel **20** in the steering process be same or different, and the control performance and the driving feeling are improved.

(75) The flow divider valve **54** may regulate the proportion of the flow of liquid flowing into the first driving device **30** from the steering control device **51** to the flow of liquid flowing into the second driving device **40** from the steering control device **51** during turning according to different vehicle speeds and a rotation direction of the steering part **52**. For example, when the vehicle speed is not more than a first preset value, the flow divider valve **54** controls the flow of liquid flowing into the first driving device **30** to be equal to the flow of liquid flowing into the second driving device **40** no matter the steering part **52** rotates forwards or reversely, thus making the deflection angles of the first directive wheel **10** and the second directive wheel **20** be equal; when the vehicle speed is more than the first preset value, the steering part **52** rotates forwards, the first directive wheel **10** located at the left side and the second directive wheel **20** located at the right side deflect leftwards, and the flow divider valve **54** controls the flow of liquid flowing into the first driving

device **30** to be less than the flow of liquid flowing into the second driving device **40**, thus making the leftward deflection angle of the first directive wheel **10** be less than leftward deflection angle of the second directive wheel **20**; when the vehicle speed is more than the first preset value, when the steering part **52** rotates reversely, the first directive wheel **10** located at the left side and the second directive wheel **20** located at the right side deflect rightwards, and the flow divider valve **54** controls the flow of liquid flowing into the second driving device **40** to be less than the flow of liquid flowing into the first driving device **30**, thus making the rightward deflection angle of the second directive wheel **20** be less than rightward deflection angle of the first directive wheel **10**.

(76) The flow divider valve **54** is provided with a liquid inlet, a first liquid outlet, and a second liquid outlet, the first liquid outlet and the second liquid outlet are communicated with the first reversing valve **511** and the second reversing valve **512** through a pipe **541** respectively, and the liquid inlet is communicated with the hydraulic pump **531** through a connecting pipe **542**. The flow divider valve **54** controls the proportion of the flow of the liquid flowing into the first driving device **30** to the flow of the liquid flowing into the second driving device **40** from the steering control device **51**, that is, controlling the proportion of the flow of the liquid discharged from the first liquid outlet to the flow of the liquid discharged from the second liquid outlet.

(77) In some embodiments, please refer to FIG. **14**, FIG. **14** is a structure diagram of a steering mechanism **100** provided by yet further embodiments of the present application. The hydraulic control system **50** further includes a flow regulating valve **55**, a vehicle speed sensor **56**, and a control unit **57**.

(78) The liquid supplied by the liquid supply device **53** flows into the first driving device **30** and the second driving device **40** through the flow regulating valve **55**, the flow divider valve **54**, and the steering control device **51** in sequence. The vehicle speed sensor **56** is configured to acquire a vehicle speed signal of the vehicle **1000**. The control unit **57** is configured to control the flow regulating valve **55** to regulate the flow of the liquid flowing into the flow divider valve **54** according to the vehicle speed signal.

(79) The vehicle speed signal of the vehicle **1000** may be acquired through the vehicle speed sensor **56**, and the flow regulating valve **55** may be controlled according to the vehicle speed signal control unit **57** to regulate the flow of the liquid flowing into the flow divider valve **54**. If the flow of the liquid flowing into the flow divider valve **54** is changed, the flow of the liquid flowing into the first driving device **30** and the second driving device **40** through the flow divider valve **54** and the steering control device **51** in sequence is also changed, thus changing the driving force provided for the first directive wheel **10** and the second directive wheel **20** by the first driving device **30** and the second driving device **40** respectively.

(80) When the vehicle speed is relatively low (the vehicle speed is not more than a second preset value, for example, the vehicle speed is not more than 30 Km/h), the vehicle speed sensor **56** acquires a vehicle speed signal representing the low vehicle speed, and if a driver turns through the steering part **52**, the flow regulating valve **55** is controlled through the control unit **57** to increase the flow of liquid flowing into the flow divider valve **54**, and the first driving device **30** and the second driving device **40** are ultimately configured to respectively provide a larger driving force for the first directive wheel **10** and the second directive wheel **20**, thus making the first directive wheel **10** and the second directive wheel **20** deflect at a relatively high speed.

(81) When the vehicle speed is relatively high (the vehicle speed is more than a second preset value, for example, the vehicle speed is more than 30 Km/h), the vehicle speed sensor **56** acquires a vehicle speed signal representing the high vehicle speed, and if a driver turns through the steering part **52**, the flow regulating valve **55** is controlled through the control unit **57** to reduce the flow of liquid flowing into the flow divider valve **54**, and the first driving device **30** and the second driving device **40** are ultimately configured to respectively provide a lesser driving force for the first directive wheel **10** and the second directive wheel **20**, thus making the first directive wheel **10** and the second directive wheel **20** deflect at a relatively slow speed, and the safety during high-speed

traveling is improved.

(82) The flow regulating valve **55** is arranged on the connecting pipe **542** connected between the hydraulic pump **531** and the liquid inlet of the flow divider valve **54**. The flow regulating valve **55** may be an electromagnetic flow regulating valve **55**.

(83) The vehicle speed sensor **56** and the flow regulating valve **55** are all electrically connected to the control unit **57**, and the control unit **57** may be an ECU (Electronic Control Unit) of the vehicle **1000**.

(84) In some embodiments, please continue to refer to FIG. **14**, the hydraulic control system **50** further includes a torque sensor **58** which is configured to acquire a torque signal of the steering part **52**. The control unit **57** is also configured to control the flow regulating valve **55** to regulate the flow of the liquid flowing into the flow divider valve **54** according to the torque signal.

(85) Where the torque sensor **58** is electrically connected to the control unit **57**.

(86) The torque signal of the steering part **52** may be acquired through the torque sensor **58**, and the flow regulating valve **55** may be controlled according to the torque signal control unit **57** to regulate the flow of the liquid flowing into the flow divider valve **54**. If the flow of the liquid flowing into the flow divider valve **54** is changed, the flow of the liquid flowing into the first driving device **30** and the second driving device **40** through the flow divider valve **54** and the steering control device **51** in sequence is also changed, thus changing the driving force provided for the first directive wheel **10** and the second directive wheel **20** by the first driving device **30** and the second driving device **40** respectively.

(87) When the torque applied to the steering part **52** by a driver in the steering process of the steering part **52** is more than a preset value, the torque sensor **58** may acquire a torque signal representing the magnitude of the torque applied to the steering part **52**, and the flow regulating valve **55** is controlled through the control unit **57** to increase the flow of liquid flowing into the flow divider valve **54**, and the first driving device **30** and the second driving device **40** are ultimately configured to respectively provide a larger driving force for the first directive wheel **10** and the second directive wheel **20**, thus making the first directive wheel **10** and the second directive wheel **20** deflect at a relatively high speed; when the torque applied to the steering part **52** by a driver in the steering process of the steering part **52** is less than the preset value, the torque sensor **58** may acquire a torque signal representing the magnitude of the torque applied to the steering part **52**, and the flow regulating valve **55** is controlled through the control unit **57** to reduce the flow of liquid flowing into the flow divider valve **54**, and the first driving device **30** and the second driving device **40** are ultimately configured to respectively provide a lesser driving force for the first directive wheel **10** and the second directive wheel **20**, thus making the first directive wheel **10** and the second directive wheel **20** deflect at a relatively slow speed.

(88) Please refer to FIG. **15**, FIG. **15** is a schematic block diagram of an apparatus **2000** for producing a steering mechanism provided by some embodiments of the present application. The apparatus **2000** for producing a steering mechanism includes a providing device **1100** and an installing device **1200**.

(89) Where the providing device **1100** is configured to provide a first directive wheel **10**, a second directive wheel **20**, a first driving device **30**, and a second driving device **40**.

(90) The installing device **1200** is configured to respectively install the first directive wheel **10** and the second directive wheel **20** on the first driving device **30** and the second driving device **40**.

(91) Where the first driving device **30** and the second driving device **40** are configured to separately drive the first directive wheel **10** and the second directive wheel **20** respectively, thus making the first directive wheel **10** and the second directive wheel **20** deflect in the same direction to achieve steering.

(92) Please refer to FIG. **16**, FIG. **16** is a flow diagram of a method for producing a steering mechanism **100** provided by some embodiments of the present application. The method for producing a steering mechanism **100** includes:

(93) **S100**: providing a first directive wheel **10**, a second directive wheel **20**, a first driving device **30**, and a second driving device **40**.

(94) **S200**: installing the first directive wheel **10** on the first driving device **30**;

(95) and **S300**: installing the second directive wheel **20** on the second driving device **40**.

(96) Where the first driving device **30** and the second driving device **40** are configured to separately drive the first directive wheel **10** and the second directive wheel **20** respectively, thus making the first directive wheel **10** and the second directive wheel **20** deflect in the same direction to achieve steering.

(97) It should be noted that step **S200** may be performed before step **S300** or may be performed after step **S300**.

(98) It should be noted that the embodiments and the features in the embodiments in the present application may be combined with each other without conflict.

(99) While the present application has been described with reference to preferred embodiments, various modifications may be made thereto and equivalents may be substituted for components thereof without departing from the scope of the present application. In particular, various technical features mentioned in the various embodiments may be combined in any way, as long as there is no structural conflict. The present application is not limited to the particular embodiments disclosed herein, but includes all technical solutions falling within the scope of the claims.

Claims

1. A steering mechanism, comprising: a first directive wheel; a second directive wheel; a first driving device connected to the first directive wheel; and a second driving device connected to the second directive wheel; wherein the first driving device and the second driving device are configured to separately drive the first directive wheel and the second directive wheel respectively, thus making the first directive wheel and the second directive wheel deflect in a same direction as each other to achieve steering; a hydraulic control system configured to supply liquid to the first driving device and the second driving device, thus making the first driving device and the second driving device separately drive the first directive wheel and the second directive wheel respectively, and making the first directive wheel and the second directive wheel deflect in the same direction; a steering control device having a first working state and a second working state; a steering part in transmission connection with the steering control device, wherein the steering part is configured to control the steering control device to switch between the first working state and the second working state; and a liquid supply device, the liquid supply device being configured to supply liquid to the first driving device and the second driving device when the steering control device is in the first working state, thus making the first driving device and the second driving device respectively drive the first directive wheel and the second directive wheel to deflect leftwards in the same direction, and being also configured to supply the liquid to the first driving device and the second driving device when the steering control device is in the second working state, thus making the first driving device and the second driving device respectively drive the first directive wheel and the second directive wheel to deflect rightwards in the same direction; a flow divider valve, wherein the liquid supplied by the liquid supply device flows into the first driving device and the second driving device through the flow divider valve and the steering control device in sequence; the flow divider valve is configured to regulate a proportion of flow of the liquid flowing into the first driving device from the steering control device to flow of the liquid flowing into the second driving device from the steering control device, thus making a deflection angle of the first directive wheel and a deflection angle of the second directive wheel be the same or different; wherein the hydraulic control system further comprises: a flow regulating valve, wherein the liquid supplied by the liquid supply device flows into the first driving device and the second driving device through the flow regulating valve, the flow divider valve, and the steering control device in sequence; a vehicle

speed sensor for acquiring a vehicle speed signal of a vehicle; and a control unit configured to control the flow regulating valve to regulate the flow of the liquid flowing into the flow divider valve according to the vehicle speed signal.

2. The steering mechanism according to claim 1, wherein the first driving device comprises: two first hydraulic cylinders distributed at a same side of the first directive wheel at an interval, wherein one of the two first hydraulic cylinders is configured to push the first directive wheel, and another first hydraulic cylinder of the two first hydraulic cylinders is configured to pull the first directive wheel, thus driving the first directive wheel to deflect.

3. The steering mechanism according to claim 2, wherein the second driving device comprises: two second hydraulic cylinders distributed at a same side of the second directive wheel at an interval, wherein one of the two second hydraulic cylinders is configured to push the second directive wheel, and another second hydraulic cylinder of the two second hydraulic cylinders is configured to pull the second directive wheel, thus driving the second directive wheel to deflect.

4. The steering mechanism according to claim 3, wherein the two first hydraulic cylinders comprise a first front hydraulic cylinder and a first rear hydraulic cylinder, the two second hydraulic cylinders comprise a second front hydraulic cylinder and a second rear hydraulic cylinder, and the first front hydraulic cylinder, the first rear hydraulic cylinder, the second front hydraulic cylinder, and the second rear hydraulic cylinder are all located between the first directive wheel and the second directive wheel; when the first directive wheel and the second directive wheel deflect leftwards in the same direction, the first front hydraulic cylinder is configured to push the first directive wheel, the first rear hydraulic cylinder is configured to pull the first directive wheel, the second front hydraulic cylinder is configured to pull the second directive wheel, and the second rear hydraulic cylinder is configured to push the second directive wheel; and when the first directive wheel and the second directive wheel deflect rightwards in the same direction, the first front hydraulic cylinder is configured to pull the first directive wheel, the first rear hydraulic cylinder is configured to push the first directive wheel, the second front hydraulic cylinder is configured to push the second directive wheel, and the second rear hydraulic cylinder is configured to pull the second directive wheel.

5. The steering mechanism according to claim 1, wherein the hydraulic control system further comprises: a torque sensor configured to acquire a torque signal of the steering part; and the control unit is also configured to control the flow regulating valve to regulate the flow of the liquid flowing into the flow divider valve according to the torque signal.

6. The steering mechanism according to claim 1, wherein the steering control device comprises: a first reversing valve connected to the first driving device through a first pipeline and a second pipeline; and a second reversing valve communicated with the second driving device through a third pipeline and a fourth pipeline; wherein the steering part is configured to switch positions of a valve core of the first reversing valve and a valve core of the second reversing valve, thus making the steering control device be in a first working state or a second working state; when the steering control device is in the first working state, the first pipeline and the third pipeline are liquid inlet pipelines, and the second pipeline and the fourth pipeline are liquid return pipelines; and when the steering control device is in the second working state, the first pipeline and the third pipeline are liquid return pipelines, and the second pipeline and the fourth pipeline are liquid inlet pipelines.

7. The steering mechanism according to claim 6, wherein the first driving device comprises: two first hydraulic cylinders distributed at a same side of the first directive wheel at an interval, wherein a piston rod of each of the two first hydraulic cylinders is connected to the first directive wheel, each of the two first hydraulic cylinders comprises a first cavity into which the liquid is fed to make the piston rod extend out and a second cavity into which the liquid is fed to make the piston rod retract; wherein the first cavity of one of the two first hydraulic cylinders and the second cavity of another first hydraulic cylinder are communicated with the first pipeline, and the second cavity of the another first hydraulic cylinder of the two first hydraulic cylinders and the first cavity of

another first hydraulic cylinder of the two first hydraulic cylinders are communicated with the second pipeline.

8. The steering mechanism according to claim 7, wherein the second driving device comprises: two second hydraulic cylinders distributed at a same side of the second directive wheel at an interval, wherein a piston rod of each of the two second hydraulic cylinders is connected to the second directive wheel, each of the two second hydraulic cylinders comprises a third cavity into which the liquid is fed to make the piston rod extend out and a fourth cavity into which the liquid is fed to make the piston rod retract; wherein the third cavity of one of the two second hydraulic cylinders and the fourth cavity of another second hydraulic cylinder of the two second hydraulic cylinders are communicated with the third pipeline, and the fourth cavity of the one of the two second hydraulic cylinders and the third cavity of another second hydraulic cylinder of the two second hydraulic cylinders are communicated with the fourth pipeline.

9. A vehicle, comprising the steering mechanism according to claim 1.

10. A steering mechanism, comprising: a first directive wheel; a second directive wheel; a first driving device connected to the first directive wheel; and a second driving device connected to the second directive wheel; wherein the first driving device and the second driving device are configured to separately drive the first directive wheel and the second directive wheel respectively, thus making the first directive wheel and the second directive wheel deflect in a same direction as each other to achieve steering; wherein the first driving device comprises: two first hydraulic cylinders distributed at a same side of the first directive wheel at an interval, wherein a piston rod of each of the two first hydraulic cylinders is connected to the first directive wheel, each of the two first hydraulic cylinders comprises a first cavity into which a liquid is fed to make the piston rod extend out and a second cavity into which the liquid is fed to make the piston rod retract; wherein the first cavity of one of the two first hydraulic cylinders and the second cavity of another first hydraulic cylinder of the two first hydraulic cylinders are communicated with a first pipeline, and the second cavity of the another first hydraulic cylinder of the two first hydraulic cylinders and the first cavity of the another first hydraulic cylinder of the two first hydraulic cylinders are communicated with a second pipeline; wherein the second driving device comprises: two second hydraulic cylinders distributed at a same side of the second directive wheel at an interval, wherein a piston rod of each of the two second hydraulic cylinders is connected to the second directive wheel, each of the two second hydraulic cylinders comprises a third cavity into which the liquid is fed to make the piston rod extend out and a fourth cavity into which the liquid is fed to make the piston rod retract; wherein the third cavity of one of the two second hydraulic cylinders and the fourth cavity of another second hydraulic cylinder of the two second hydraulics cylinders are communicated with a third pipeline, and the fourth cavity of the one of the two second hydraulic cylinders and the third cavity of the another second hydraulic cylinder of the two second hydraulics cylinders are communicated with a fourth pipeline.
